

Grade Eight Science - Body Systems Unit 1

MYP Criterion A Practice Questions

- This document contains twelve practice questions for the upcoming Body Systems assessment
- All questions are based on MYP Criterion A (See below)
- There are three practice questions from each level; one for each strand
- The MYP Command Terms used in the questions are provided below

MYP Science Assessment Criterion A: Knowing and Understanding

Achievement Level	Level Descriptor
0	The student does not reach a standard indicated by any of the descriptors below
1-2	The student is able to: • recall scientific knowledge • apply scientific knowledge and understanding to suggest solutions to problems set in familiar situations • apply information to make judgments
3-4	The student is able to: • state scientific knowledge • apply scientific knowledge and understanding to solve problems set in familiar situations • apply information to make scientifically supported judgments.
5-6	The student is able to: • outline scientific knowledge • apply scientific knowledge and understanding to solve problems set in familiar situations and suggest solutions to problems set in unfamiliar situations • interpret information to make scientifically supported judgments
7-8	The student is able to: • describe scientific knowledge • apply scientific knowledge and understanding to solve problems set in familiar and unfamiliar situations • analyze information to make scientifically supported judgments

MYP Command Terms Used in the Questions

State: Give a specific name, value, or other brief answer without explanation of calculator	Outline: Give a brief account or summary
Apply: Use knowledge and understanding in response to a given situation	Describe: Give a detailed account or picture of a situation, event, pattern, or process
Recall: Remember or recognize from prior learning experience	Suggest: Propose a solution, hypothesis, or other possible answer
Analyze: To identify parts and relationships, and to interpret information to reach conclusions	Interpret: Use knowledge and understanding to recognize trends and draw conclusions from given information

LEVELLED BODY SYSTEMS PRACTICE QUESTIONS - CRITERION A

Level 1-2 Practice Questions

1. **Recall** that the nervous system is made up of the:
- A. Heart, brain, and spinal cord
 - B. **Nerves, brain, and spinal cord**
 - C. Nerves, brain, spinal cord, and heart
 - D. Nerves and Brain
2. A person went to the hospital with trouble breathing. **Apply** scientific knowledge and understanding to **suggest** which part of the human anatomy, from the list below, is likely damaged.
- A. **Trachea**
 - B. Esophagus
 - C. Stomach
 - D. Pancreas
 - E. None of the above
 - F. All of the above
3. A student left school with stomach pain. When examined, the doctor found an ulcer. This means that acid burned through the stomach lining. **Apply** information to choose which item, from the list below, likely caused the ulcer. Was it caused by:
- A. Folds in the stomach
 - B. Peristalsis
 - C. Valves not working properly
 - D. **Not enough mucus**

Level 3-4 Practice Questions

1. **State** the difference between an artery and a vein

An artery takes blood away from the heart and a vein takes blood to the heart

2. During a lesson on the nervous system, a student hits their knee with a reflex hammer. They were surprised when nothing happened, as they expected their leg to lift upward. **Apply** your scientific knowledge and understanding to help the student understand why they did not get the result they expected.

When you tap the knee in the correct spot below the knee cap the leg will move upward. This is because you hit a sensory receptor which is part of a nerve. If the leg does not move then you have tapped the leg in the wrong spot and you have not hit the sensory receptor on the nerve in that location.

Level 5-6 Practice Questions:

1. **Apply** information learned in class to make a scientifically supported judgement about why the intestines are the longest part of the human digestive system

The human body has two intestines: small and large (aka: colon). The intestines together make up the longest part of the human digestive system because they do two important jobs.

1. The small intestine must be really long, about 7 m, because the food (or what's left of it) must stay in there long enough for most of the nutrients to be absorbed.
2. The large intestine (or colon), although shorter than the small intestine, removes waste from our bodies and the waste must stay in there long enough for enough water to be absorbed from the waste.

2. At high altitudes the air is said to be "thin," meaning that it has less oxygen. **Apply** scientific knowledge and understanding to **suggest one way** the human body could be adapted to make living at high altitudes easier

Sample Answer:

If we were able to adapt the human body to live at high altitudes a few things we could do are suggested below:

1. Bigger lungs - bigger lungs would allow more oxygen to enter the respiratory system, therefore, providing more oxygen for the entire body
2. Increasing the number of red blood cells - red blood cells carry oxygen around the body, so more red blood cells means more oxygen could be moved

3. A coach tells a team of Sprinters that, because they want to leave the start as fast as possible, they should rely on visual cues, rather than sound cues at the start of a race.

Table 1. Reaction Times of Different Stimulus

Type of Stimulus	Reaction Time (milliseconds)
Visual (Seeing a light)	250
Auditory (Hearing a beep)	270
Tactile (feeling a tap)	200

Interpret data from Table 1 above to make a scientifically supported judgement about:

- A. **Describe** the body system that the coach telling the Sprinters to use at the start of the race
- B. If the coach is giving information that makes sense to the Sprinters

Sample Answer:

A. By telling the sprinters to rely on various "cues," or stimuli, the coach is

telling the Sprinters to use the nervous system. The nervous system is the system where the brain sends signals down the spinal cord to nerves in different parts of the body. For a visual stimulus, like seeing light (Table 1), the brain sends a signal to a nerve behind the eye.

B. Since the job of a sprinter is to sprint the fastest around the route of the race, they want to leave the starting point as fast as possible. Therefore, they are best to use the type of stimulus that has the fastest reaction time as their cue to start.

The coach told the Sprinters to use "auditory," cues, with a reaction time of 270 ms. Even though the reaction time is slower, sound might be less distracting to the Sprinters than a visual cue, even though it has a faster reaction time of 250 ms. The tactile cue has the lowest reaction time of 200 ms, but since this requires touch it would likely not be possible to do this at the start of the race.

Level 7-8 Practice Questions:

1. **Describe** scientific knowledge by giving two similarities and two differences between the Digestive and Circulatory systems

Two Similarities (Sample Answer)

Similarity Number One	Similarity Number Two
Both systems use diffusion: the movement of molecules from an area of high concentration to an area of low concentration. The digestive system diffuses nutrients into the bloodstream, while, the circulatory system diffuses carbon dioxide into the respiratory system	Both systems deliver necessary molecules of life. The digestive system delivers nutrients to our organs and the circulatory system delivers oxygen to our organs. Both are needed for organs to stay alive.

Two Differences (Sample Answer)

Difference Number One	Difference Number Two
The digestive system follows a one way path - food enters at the mouth and waste exits at the anus Rather than a one way path,the circulatory system is a connected circuit. Blood circulates through the heart and blood vessels changing from oxygenated to deoxygenated	The waste products in each system are different. The digestive system processes food, removes the nutrients, and leaves a solid waste product called stool (aka. poop). In the circulatory system the waste product is a gas called carbon dioxide. It is produced in the bloodstream by cellular respiration and it leaves the body via diffusion from the alveoli and exhalation

2. Smoking is a global problem that impacts different body systems. Smoking is a process where chemicals are inhaled into the lungs. Smoking can cause lung and pharynx cancer and it makes veins and arteries become more narrow. It can also cause breathing problems.

After reading the paragraph above, **apply** scientific knowledge and understanding to decide which three body systems studied in class are impacted by smoking. Use specific pieces of information from the paragraph to support your answer.

Body System	Specific Information to Support
1. Respiratory System - paragraph says chemicals are inhaled into the lungs and that smoking can cause lung and pharynx cancer and breathing problems 2. Circulatory System - paragraph says smoking can make blood vessels become more narrow. 3. Digestive System - paragraph says smoking can cause pharynx cancer. The pharynx is an anatomical part in common between the digestive and respiratory systems	

3. **Analyze** information from the paragraph in question 2 above to decide
- A. What factor might be causing veins and arteries to become more narrow in smokers
 - B. Using scientific knowledge learned in class **describe** one problem that you think could be caused by narrow veins and arteries

A. I think that it's possible that the narrowing of veins and arteries could be caused by the chemicals people are breathing in while smoking. The paragraph says that smoking is the process of taking chemicals into the lungs, so I think this makes sense.

B. The job of blood vessels is to carry blood around the body, and because oxygen travels in the bloodstream, the job of blood vessels is also to carry oxygen. If blood vessels become more narrow it's going to be more difficult for blood to move through them. Just think of how water moves through a clear pipe, versus one that has stuff clogging its opening.

If blood has trouble moving through the bloodstream, then oxygen is also not getting to the organs in an efficient manner. This is a problem because organs rely on oxygen for survival and they will not work properly without sufficient.